SIMATS ENGINEERING
ASSESSMENT TOOL 1

NAME: N.Tejaswitha **REG.NO:** 192371057
COURSE NAME: CSA16Data Warehousing and Data Mining **COURSE CODE:** CSA1603

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:50**

Annexure A

Data Interpretation

Criteria	Understanding of Data ***(2.5)***	Analysis ***(2.5)***	Application of Concepts ***(2)***	Conclusion ***(1.5)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I **_N.Tejaswitha_ / *192371057*__***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

N.Tejaswitha
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q1 Dataset: Hospital patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values:

K	SSE
1	600
2	350
3	220
4	200
5	190

Question:

A health-care organisation uses K-means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting the number of clusters affects patient segmentation and healthcare decision-making.

SSE-Means → Sum of Squared Errors.

- We are Using Elbow Method
- The reduction in SSE in

K	SSE	Reduction from Previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

We Observe that when

$K = 2$ & 3 the SSE value is been reduced much and when K value is 4 & 5 the decrease is very small. This variation happens after the value of K is 3. Therefore Elbow method suggest the value of K is 3.

∴ Based on health risk score 3 clusters have formed.

(i) Low risk - with low Expense.

(ii) Moderate risk - with moderate Expense.

(iii) High risk - with high Expense.

* Impact on Decision Making:

⇒ For Low risk group:

(i) Preventive care

(ii) Routine monitoring.

⇒ For moderate risk:

(i) Regular follow-up

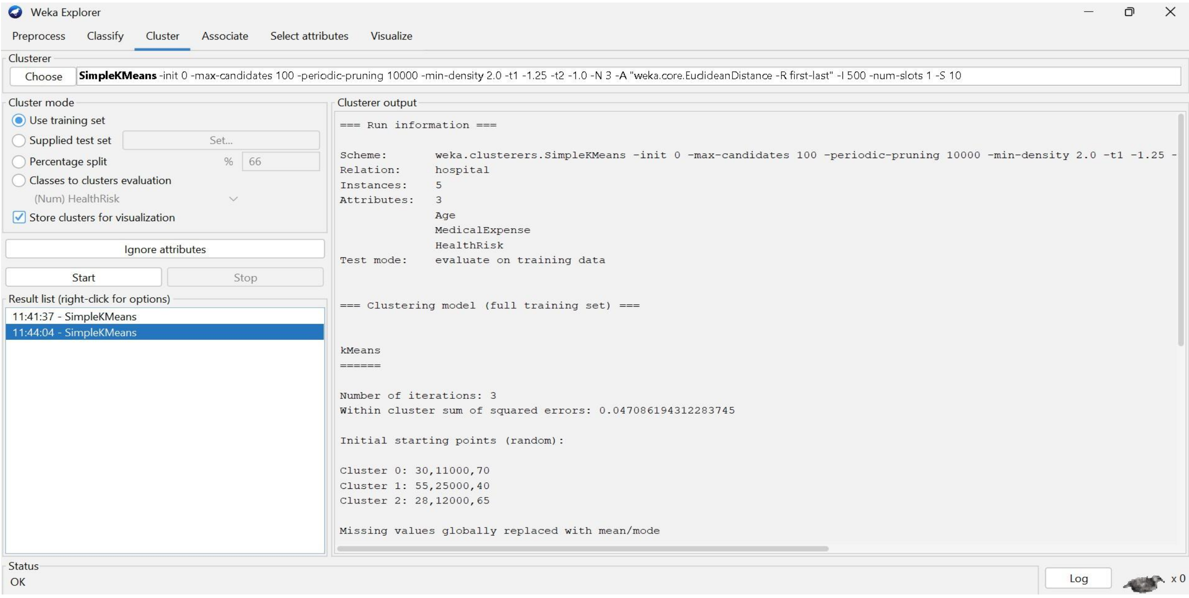
(ii) Targeted intervention.

⇒ For High risk:

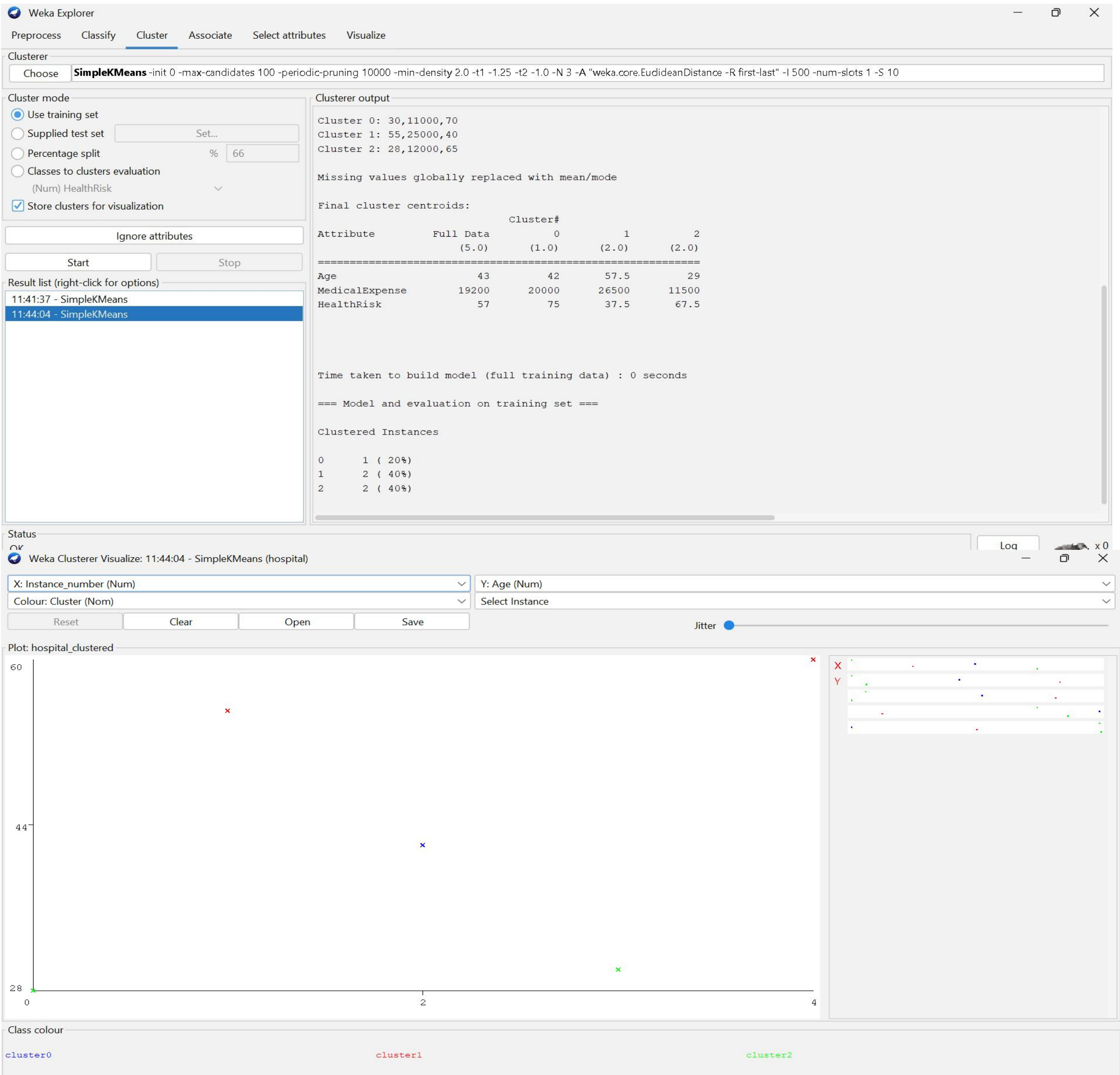
(i) Immediation intervention.

(ii) Intensive monitoring / close supervision.

OUT PUT :



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRisk
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
11:41:37 - SimpleKMeans
11:44:04 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation: hospital
Instances: 5
Attributes: 3
Age
MedicalExpense
HealthRisk
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 3
Within cluster sum of squared errors: 0.047086194312283745
Initial starting points (random):
Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,65
Missing values globally replaced with mean/mode
Status
OK
Log
x 0



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRisk
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
11:41:37 - SimpleKMeans
11:44:04 - SimpleKMeans
Clusterer output
Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,65
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1 2
(5.0) (1.0) (2.0) (2.0)
Age 43 42 57.5 29
MedicalExpense 19200 20000 26500 11500
HealthRisk 57 75 37.5 67.5
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 2 (40%)
2 2 (40%)
Status
Log
x 0
Weka Clusterer Visualize: 11:44:04 - SimpleKMeans (hospital)
X: Instance_number (Num)
Y: Age (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: hospital_clustered
60
44
28
0
2
4
X
Y
Class colour
cluster0
cluster1
cluster2

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior

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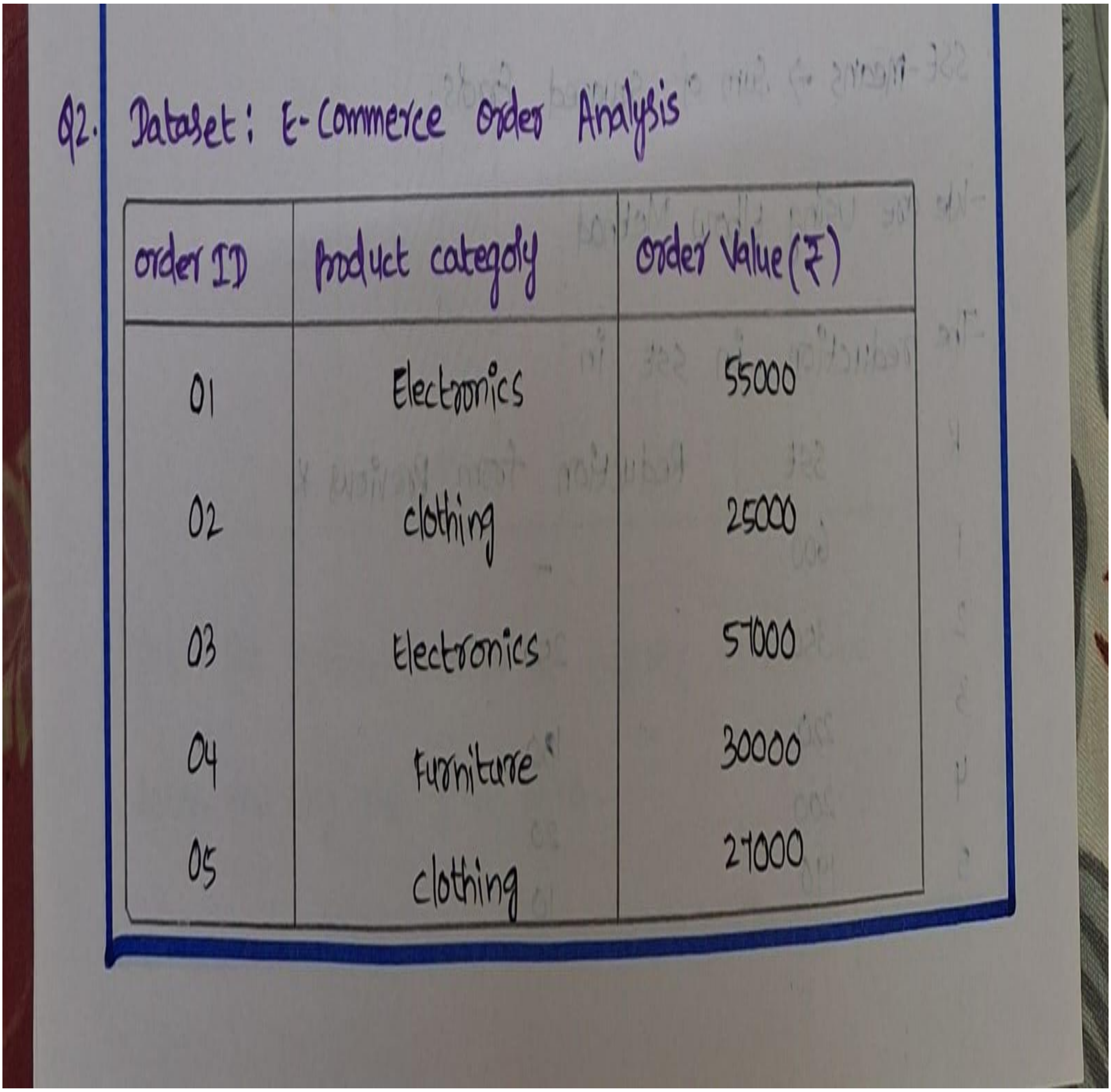


Q2. Dataset: E-Commerce Order Analysis

order ID	product category	order value (₹)
01	Electronics	55000
02	clothing	25000
03	Electronics	57000
04	Furniture	30000
05	clothing	27000

Question :

An online Retailer applies hierarchial clustering to group customer orders based on Product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

HAC :

⇒ It calculates the distance between the clusters

⇒ Merges the two closest clusters.

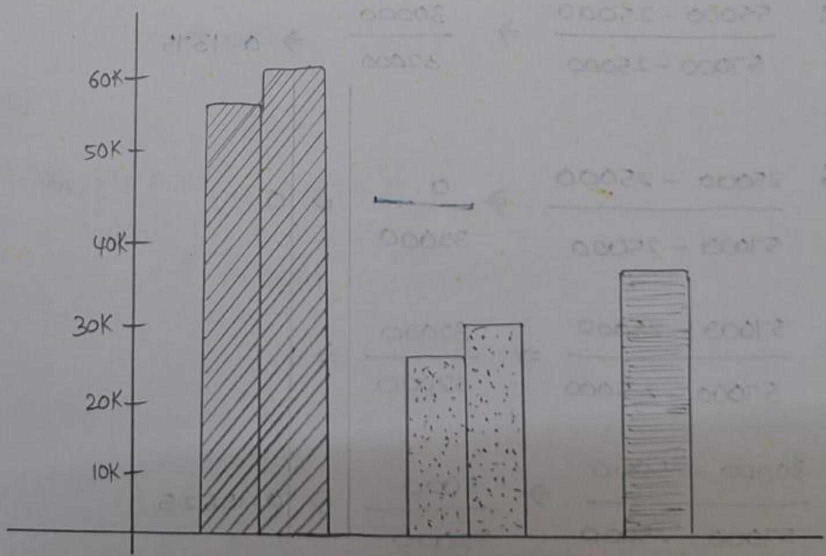
⇒ Recalculates the distance.

⇒ continue until it will become as one cluster.

⇒ O_1 & O_3 forms 1st cluster - Electronics.

⇒ O_2 & O_5 forms 2nd cluster - clothing

⇒ O_4 forms 3rd cluster - furniture.



But the optimal number of clusters cannot be determine by the given data.

Step-01 : Use one-hot Encoding

order	Electronics	clothing	Furniture
O1	1	0	0
O2	0	1	0
O3	1	0	0
O4	0	0	1
O5	0	1	0

Step-02 : Normalize order value

Use Min-Max normalization

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Here;

$x_{min} = 25,000$

$x_{max} = 57,000$

$\therefore O_1 : \frac{55000 - 25000}{57000 - 25000} \Rightarrow \frac{30000}{32000} \Rightarrow 0.9375$

$O_2 : \frac{25000 - 25000}{57000 - 25000} \Rightarrow \frac{0}{32000} \Rightarrow 0$

$O_3 : \frac{57000 - 25000}{57000 - 25000} \Rightarrow \frac{32000}{32000} \Rightarrow 1$

$O_4 : \frac{30000 - 25000}{57000 - 25000} \Rightarrow \frac{5000}{32000} \Rightarrow 0.15625$

$O_5 : \frac{27000 - 25000}{57000 - 25000} \Rightarrow \frac{2000}{32000} \Rightarrow 0.0625$

order	Electronics	clothing	Furniture	Normalized Value
O_1	1	0	0	0.9375
O_2	0	1	0	0
O_3	1	0	0	1
O_4	0	0	1	0.15625
O_5	0	1	0	0.0625

Step-03: calculate Euclidean Distances

- For two observations:

$$d(P,Q) = \sqrt{\sum (P_i - Q_i)^2}$$

- O_1 & O_3

$$d(O_1, O_3) = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2}$$

$$= \sqrt{0+0+0+(-0.0625)^2}$$

$$= \sqrt{(0.00390625)}$$

$$= 0.0625$$

- O_2 & O_5

$$d(O_2, O_5) = |(0 - 0.0625)|$$

$$= 0.0625$$

- O_4

$$d(O_4) = 0.0625$$

∴ The Elements within clusters are small, so it will be in small cluster only.

step-04 : Distance Matrix

Using the same matrix of notation we are going to find distance matrix.

ID	O1	O2	O3	O4	O5
O1	0	1.697	0.063	1.616	1.663
O2	1.697	0	1.732	1.423	0.063
O3	0.063	1.732	0	1.647	1.697
O4	1.616	1.423	1.647	0	1.1417
O5	1.663	0.063	1.697	1.417	0

step-05 : Hierarchial Agglomerative clustering

We use single linkage where:

$$D(A,B) = \min d(x,y)$$

between members of the two clusters

Merge 1

smallest distance : $O1 - O3 \Rightarrow 0.0625$

By seeing smallest distance

$C_1 = O1 + O3$; $C_1 = \{O1, O3\} \Rightarrow 0.0625$

$C_2 = O_2 + O_5$; $C_2 = \{O2, O5\} \Rightarrow 0.0625$

$C_3 = O4$; $C_3 = \{O4\} \Rightarrow 0.0625$

Merging of cluster can be done by finding the distance b/w C_1 & O4 ; C_2 & O4. Minimum of C_1 & O4 , C_2 & O4

C_1 O4

(O_1, O_3) O4

$\min((O_1,O4),(O_3,O4))$

$\min(1.616, 01.647)$

↳ 1.616

C_2 O4

(O_1, O_5) O4

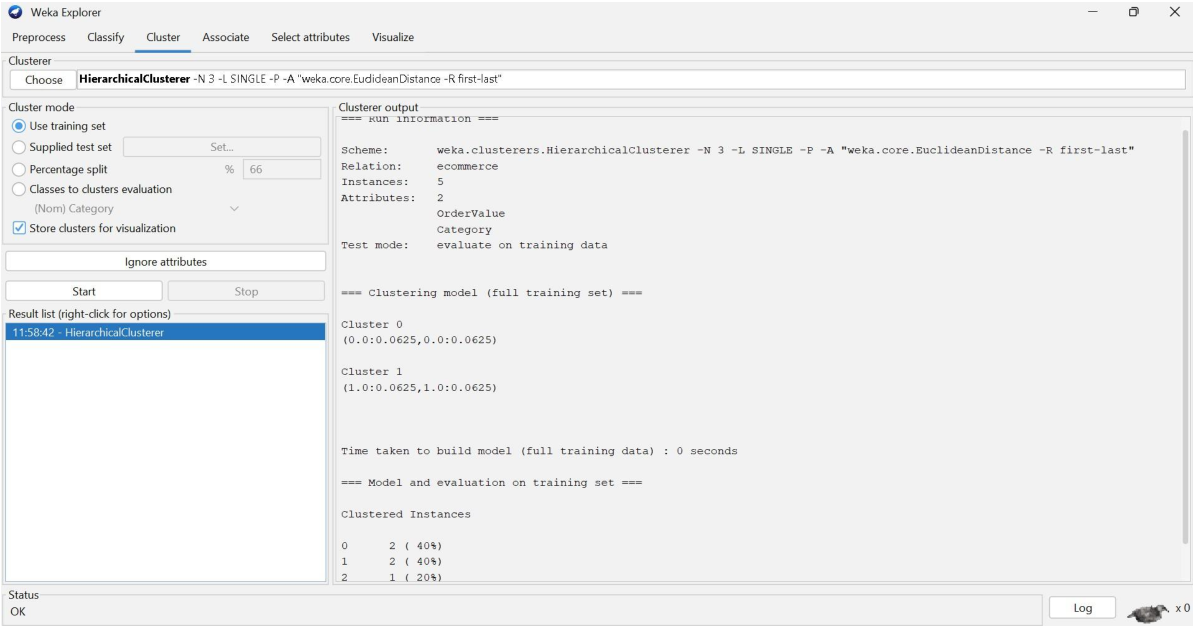
$\min((O2,O4),(O5,O4))$

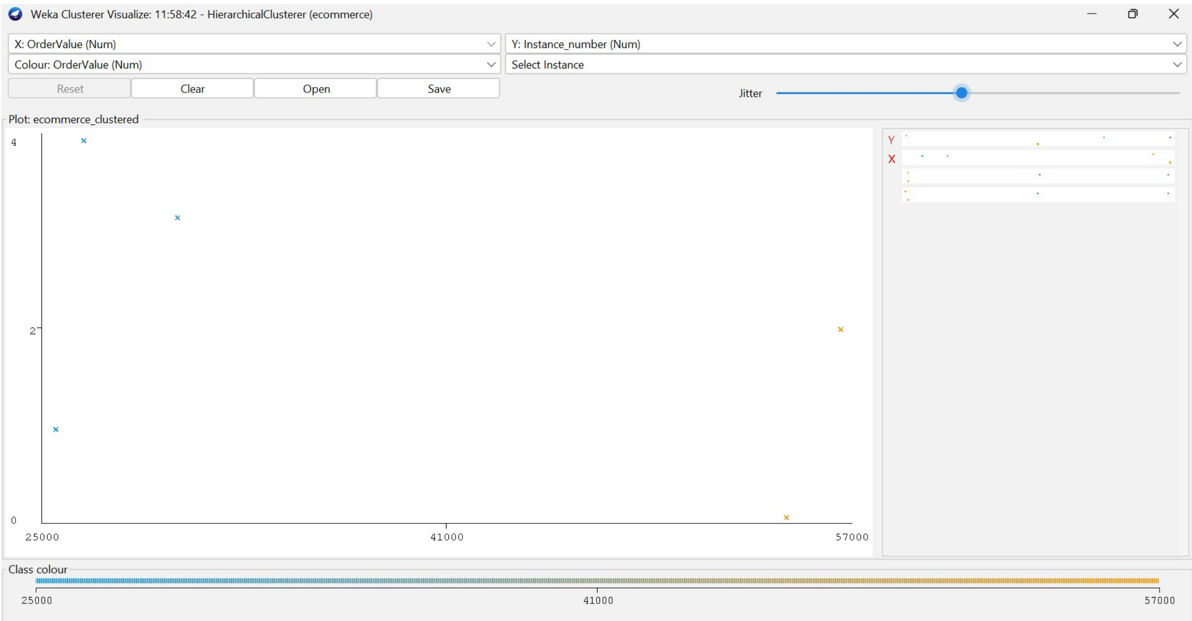
$\min(1.432, 1.417)$

↳ 1.417

O_1 $O3$ → C_1 ; O_2 O_5 → C_2 ; O4 → C_3

OUTPUT :





Q3. Dataset: Mobile App User Behaviour

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q3. Dataset: Mobile App User Behaviour.

User	Daily Usage Time (min)	Login Frequency	In-App Purchase(₹)
U_1	120	18	2000
U_2	60	5	5000
U_3	140	22	1500
U_4	50	4	6000
U_5	130	20	1800

Question:

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Step-01: Identify the features.

We have 3 features for each user:

- Usage Time (minutes)
- Login Frequency.
- Purchase Amount (₹)

Step-02: Observe the data

Look at the user's behavior.

U_1, U_3, U_5:

- High Usage
- High Login Frequency
- Lower purchase amount.

So they form one similar group

U_2, U_4:

- Low Usage

- Low Login Frequency.
- High purchase amount.

So they form another similar group.

Step-03: Compare the two groups

Features	$G_1: U_1, U_3, U_5$	$G_2: U_2, U_4$
Usage	High	Low
Logins	High	Low
Purchase	Low	High

There is a clear difference between the two groups.

Step-04: Determine No. of clusters

Since there are two distinct behavioral patterns we choose: $K = 2$

Therefore:

No. of clusters = 2

Step-05: Name the clusters

cluster 01 - Highly Engaged, lower-purchase users.
→ U_1, U_3, U_5

cluster 02 - Less Engaged, high-purchase users.
→ U_2, U_4

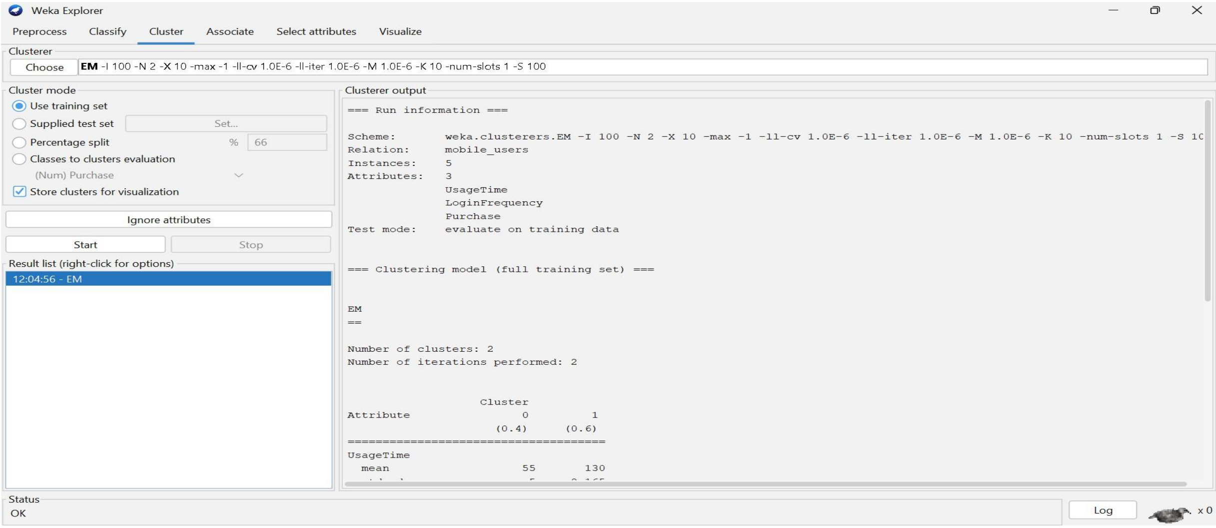
Step-06: Final Results

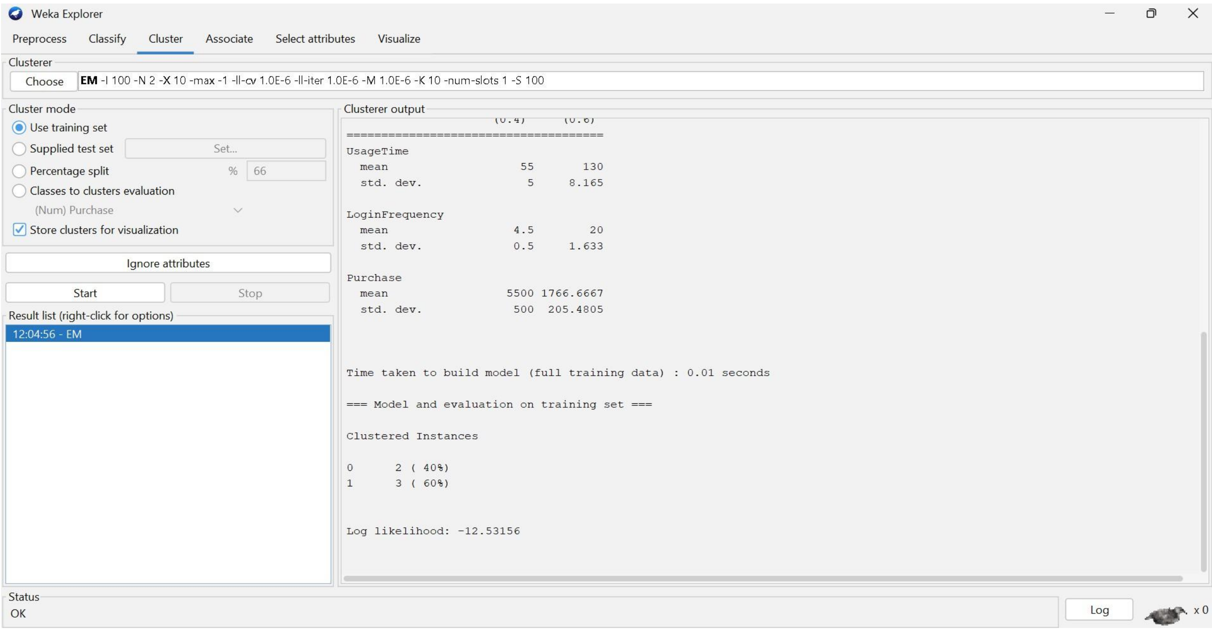
$K = 2$

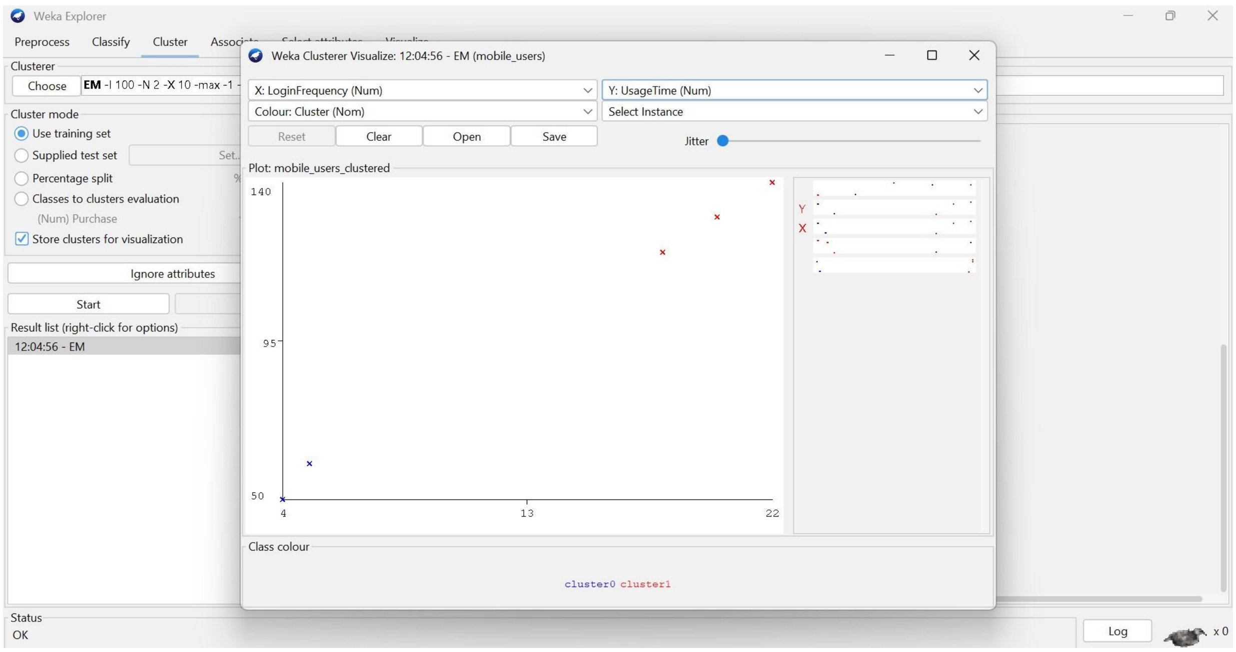
cluster 1 = $\{U_1, U_3, U_5\}$.
cluster 2 = $\{U_2, U_4\}$.

I analyzed three features - usage, log in, frequency, and purchase amount. The users show two clearly different behavioural patterns. ∴ The appropriate no. of clusters formed is "2".

OUTPUT:


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose EM -I 100 -N 2 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set Set...
Percentage split % 66
Classes to clusters evaluation
(Num) Purchase
Store clusters for visualization
Ignore attributes
Start Stop
Result list (right-click for options)
12:04:56 - EM
Clusterer output
=== Run information ===
Scheme: weka.clusterers.EM -I 100 -N 2 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 10
Relation: mobile_users
Instances: 5
Attributes: 3
UsageTime
LoginFrequency
Purchase
Test mode: evaluate on training data
=== Clustering model (full training set) ===
EM
==
Number of clusters: 2
Number of iterations performed: 2
Cluster
Attribute 0 1
(0.4) (0.6)
UsageTime
mean 55 130
Status
OK
Log x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose EM -I 100 -N 2 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set Set...
Percentage split % 66
Classes to clusters evaluation
(Num) Purchase
Store clusters for visualization
Ignore attributes
Start Stop
Result list (right-click for options)
12:04:56 - EM
Clusterer output
UsageTime
mean 55 130
std. dev. 5 8.165
LoginFrequency
mean 4.5 20
std. dev. 0.5 1.633
Purchase
mean 5500 1766.6667
std. dev. 500 205.4805
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Log likelihood: -12.53156
Status
OK
Log x 0


Weka Explorer
Weka Clusterer Visualize: 12:04:56 - EM (mobile_users)
X: LoginFrequency (Num)
Y: UsageTime (Num)
Colour: Cluster (Nom)
Select Instance
Reset Clear Open Save
Jitter
Plot: mobile_users_clustered
140
95
50
4
13
22
Class colour
cluster0 cluster1
Status
OK
Log x 0

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how Normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q4. Dataset : Agricultural crop monitoring.

Farm	Rainfall (mm)	soil Fertility Score	crop yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question :

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation & improves the reliability of agricultural data analysis.

Step-01: calculate distances before normalization.

We use Euclidean distance.

$$d(P,Q) = \sqrt{\Sigma_i (P_i - Q_i)^2}$$

For example;

$$d(F_1, F_4) = \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2}$$

$$= \sqrt{400+1+25}$$

$$= \sqrt{426} \approx 20.64$$

Similarly, the important distances are :

Pair	Distance
F1 – F4	20.64

$F_2 - F_5$	20.64
$F_1 - F_3$	100.14
$F_2 - F_3$	100.52
$F_3 - F_4$	120.45
$F_3 - F_5$	120.97
$F_1 - F_2$	200.60
$F_4 - F_5$	241.37

HAC before Normalization

Using Single Linkage; the smallest distance is:

$F_1 - F_4 = 20.64$

&

$F_2 - F_5 = 20.64$

Therefore;

$C_1 = \{F_1, F_4\}$

$C_2 = \{F_2, F_5\}$

$C_3 = \{F_3\}$

Next, F_3 Joins

C_1, because:

$\min[d(F_3, F_1), d(F_3, F_4)] = \min(100.14, 120.45)$

$= 100.14$

So:

$C_1 = \{F_1, F_3, F_4\}$

Finally, F_3 connects to C_2 at 100.52, Producing one cluster.

Step-02: Min-Max Normalization.

The Min-Max formula is:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

- Rainfall : $x_{min} = 280$, $x_{max} = 520$
- soil fertility : $x_{min} = 3$, $x_{max} = 9$
- crop yield : $x_{min} = 65$, $x_{max} = 90$

Farm	Rainfall	Soil fertility	crop yield
F_1	0.0833	0.8333	0.8000
F_2	0.9167	0.1667	0.2000
F_3	0.5000	0.5000	0.6000
F_4	0.0000	1.0000	1.0000
F_5	1.0000	0.0000	0.0000

step-03: Euclidean distance

Pair	Distance
$F_1 - F_4$	0.2734
$F_2 - F_5$	0.2734
$F_1 - F_3$	0.5698
$F_2 - F_3$	0.6669
$F_3 - F_4$	0.8124
$F_3 - F_5$	0.9274

The smallest distance are $F_1 - F_4$ and $F_2 - F_5$

step-04: HAC clustering

At 3 clusters:

$C_1 = \{F_1, F_4\}$

$C_2 = \{F_2, F_5\}$

$C_3 = \{F_3\}$

∴ Thus similar farms are grouped together.

Step-05: Interpretation

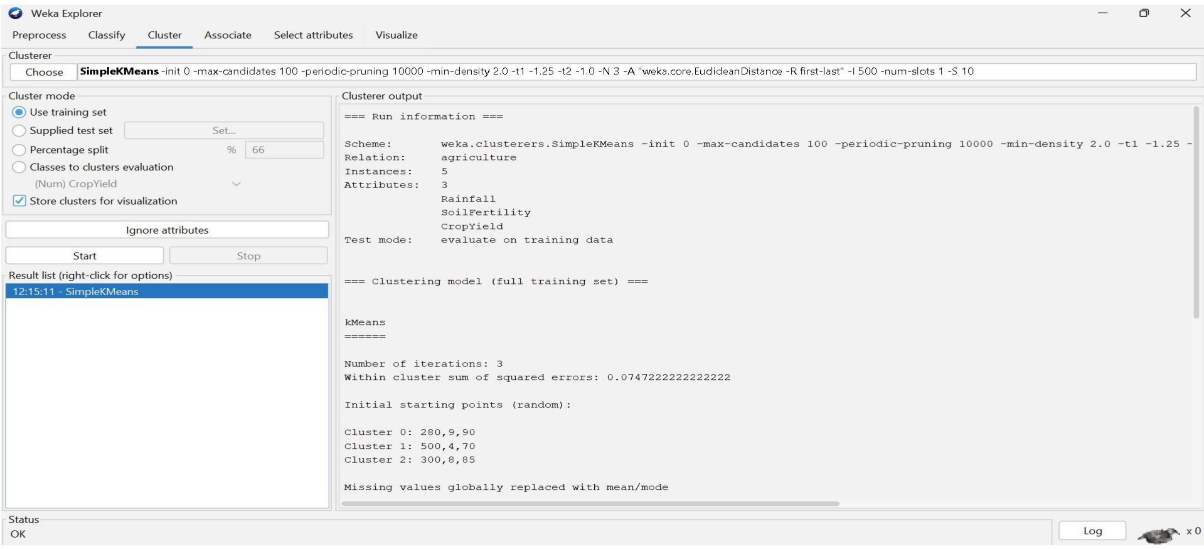
Before normalization, rainfall dominates all because its numerical range is much larger.

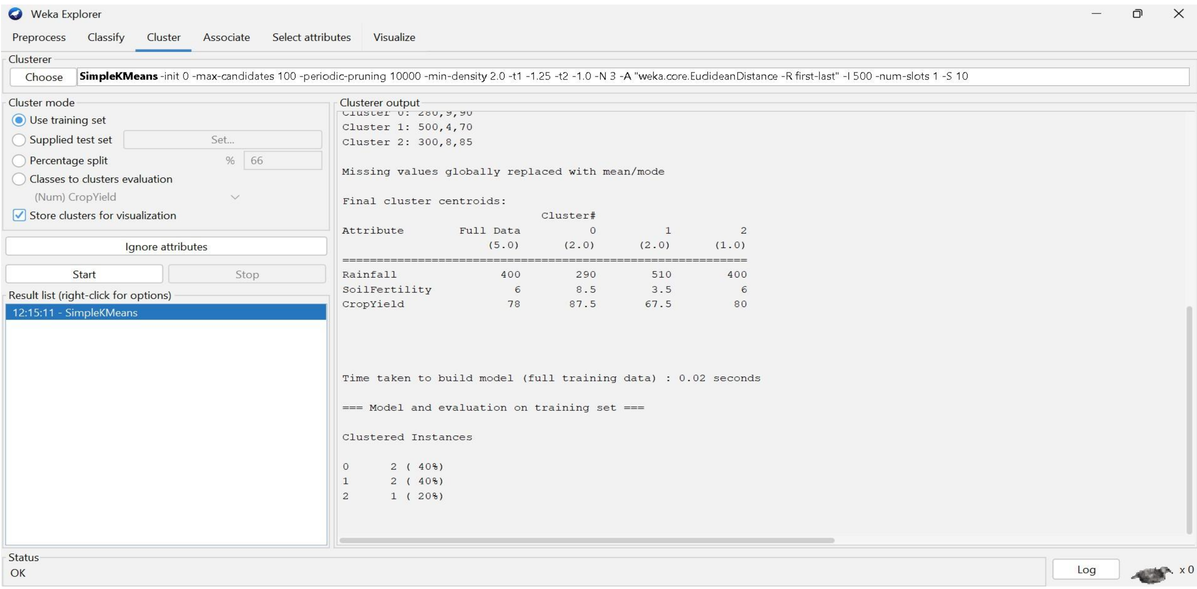
After normalization, all features are scaled to 0-1, giving rainfall, soil fertility, & crop yield to more balanced influence.

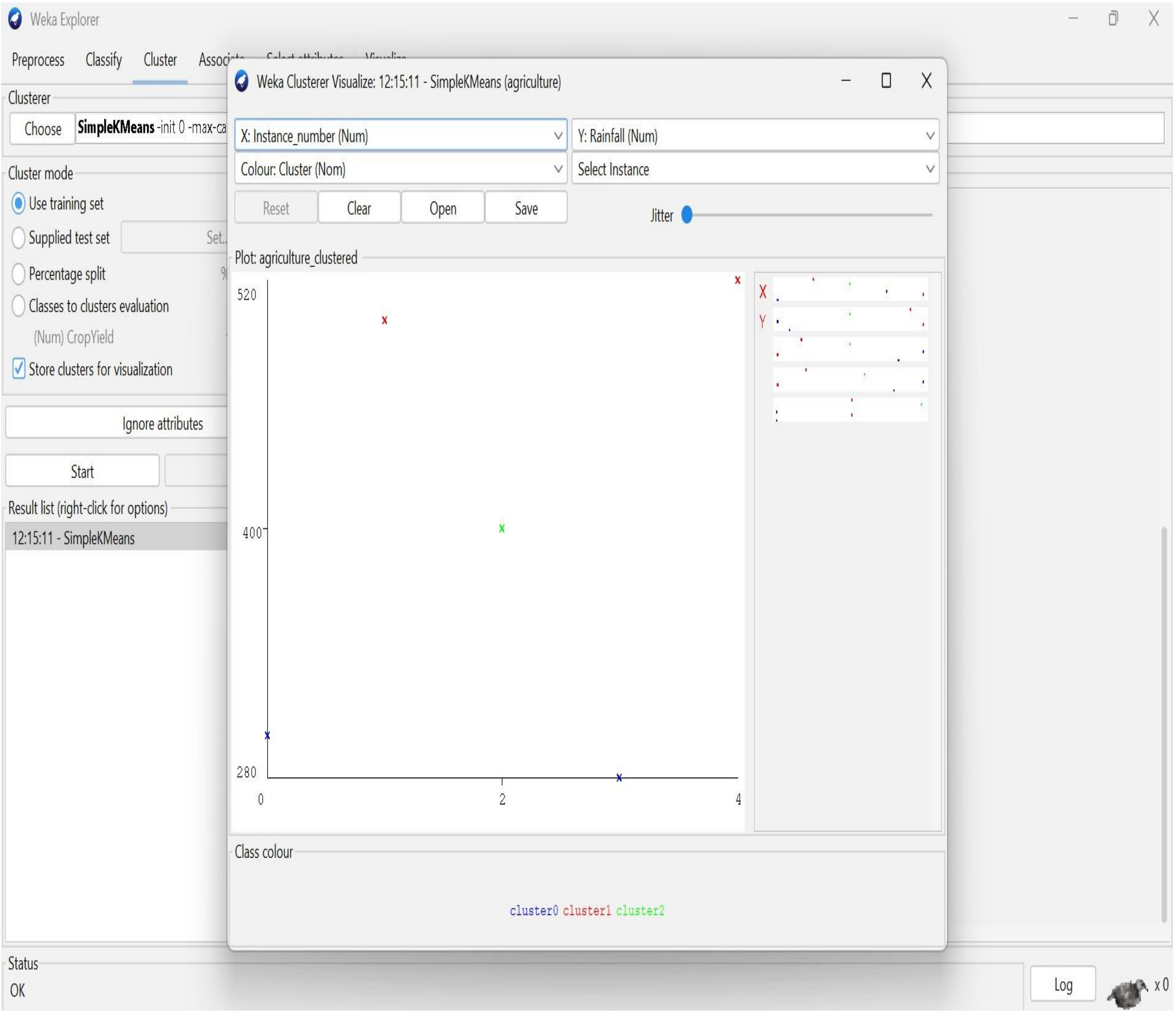
∴ The final 3 clusters:

$\{F_1, F_4\}, \{F_2, F_5\}, \{F_3\}$

OUTPUT:







Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Q5. Dataset: Employee Performance Evaluation.

Employee	Productivity score	Attendance (%)	Skill Assessment Score
E_1	88	92	84
E_2	62	95	68
E_3	78	88	72
E_4	55	65	58
E_5	93	96	90

Silhouette Scores

Model score

K=2 0.48

K=3 0.68

K=4 0.52

Question:

A company clusters Employees based on Productivity, attendance, and skill assessment scores. Analyze the

Silhouette scores obtained for different clustering models & determine the most suitable no. of clusters. Justify your answer & Explain how the selected model contributes to Effective Employee Performance Evaluation.

Given silhouette scores:

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

* Analysis :

The silhouette score measures the quality of clustering. A higher score indicates better-defined and well-separated clusters.

- K=2 : Score =0.48
  ↳ Moderate clustering.
- K=3 : Score = 0.68
  ↳ Best clustering.
- K=4 : Score = 0.52
  ↳ Moderate clustering.

Since K=3, has the highest silhouette score (0.68), it is the most suitable model.

Justification :

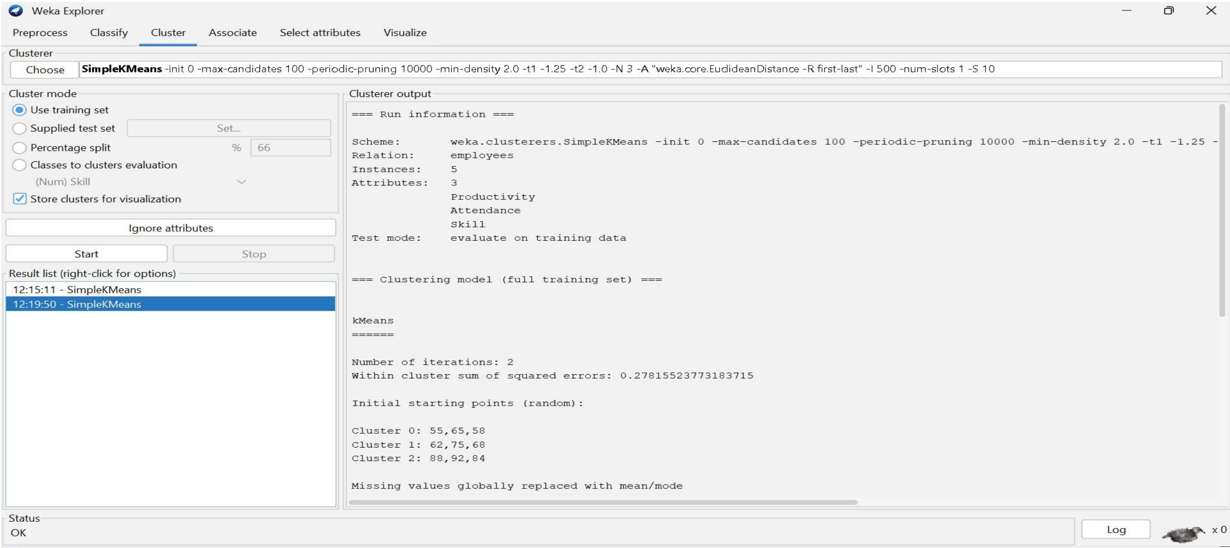
K=3 provides better similarity among employees within the same cluster and better separation between different clusters. It can Effectively group Employees based on their Productivity, attendance, & skill assessment.

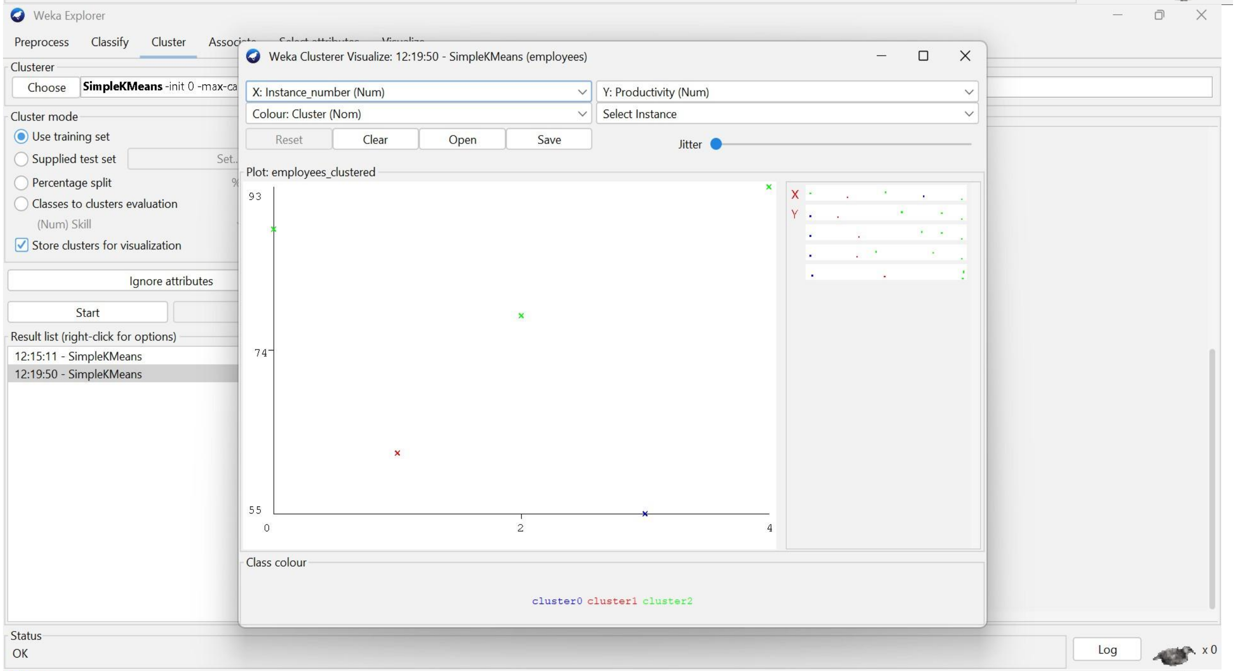
* Conclusion :

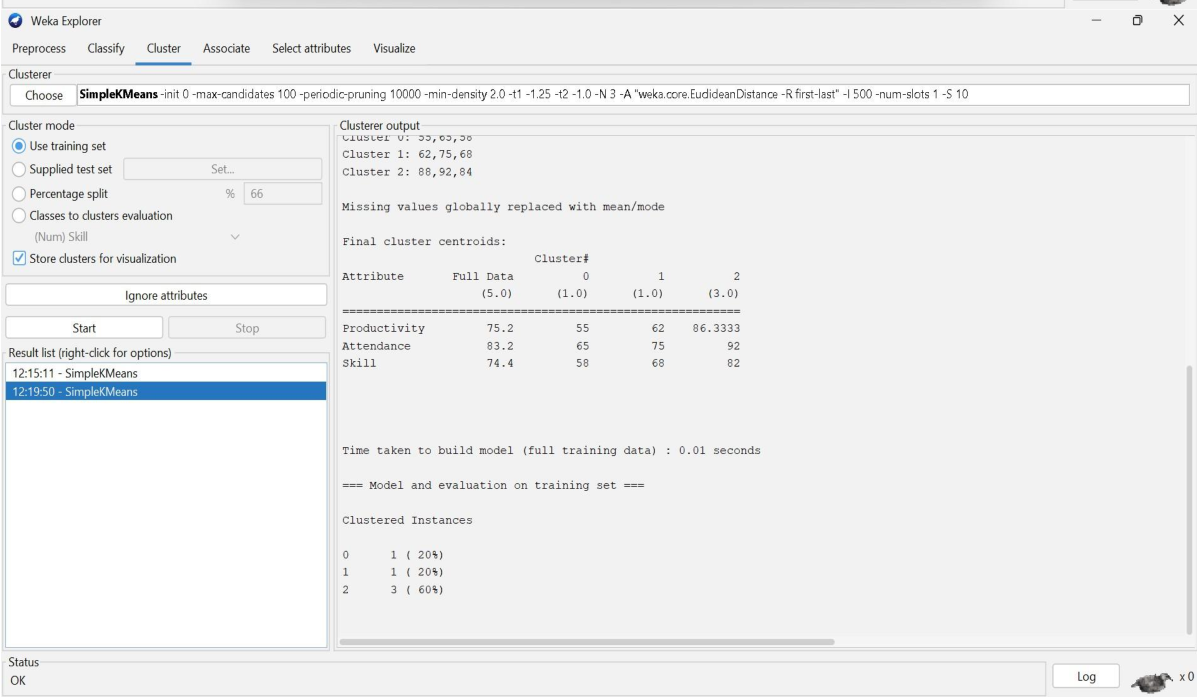
Best Number of clusters = 3

The selected K=3 model can help the company identify different employee performance groups (such as high, medium, and low performers). This supports better performance evaluation, training, recognition and employee development decisions.

OUTPUT:


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split % 66
Classes to clusters evaluation
(Num) Skill
Store clusters for visualization
Ignore attributes
Start Stop
Result list (right-click for options)
12:15:11 - SimpleKMeans
12:19:50 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation: employees
Instances: 5
Attributes: 3
Productivity
Attendance
Skill
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 2
Within cluster sum of squared errors: 0.27815523773183715
Initial starting points (random):
Cluster 0: 55,65,58
Cluster 1: 62,75,68
Cluster 2: 88,92,84
Missing values globally replaced with mean/mode
Status
OK
Log
x 0


Weka Clusterer Visualize: 12:19:50 - SimpleKMeans (employees)
X: Instance_number (Num)
Y: Productivity (Num)
Colour: Cluster (Nom)
Select Instance
Reset Clear Open Save
Jitter
Plot: employees_clustered
93
74
55
0
2
4
X
Y
Class colour
cluster0 cluster1 cluster2


Weka Explorer
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Clusterer output
Cluster 1: 62,75,68
Cluster 2: 88,92,84
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1 2
(5.0) (1.0) (1.0) (3.0)
Productivity 75.2 55 62 86.3333
Attendance 83.2 65 75 92
Skill 74.4 58 68 82
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 1 (20%)
2 3 (60%)
Status
OK